

Screener Factor Study -- Phase 2

Part 1 -- Pre-registration

*Written and committed 2026-09-14, before any factor result was computed. The decision

rules below

Question

Which signals should the screener's composite use, with what weights -- and is option 3

(timing-based)

Signals tested

Current composite components -- computed exactly as screener.py does:

Signal	Definition	Direction
mom20	close / close 19 sessions earlier ? 1	higher ranks better
vol_ratio	today's volume / 20-session average volume	higher
squeeze	Bollinger width = 4 x 20-session ? / 20-session SMA	lower ranks better
composite	0.40 rank(mom20) + 0.30 rank(vol_ratio) + 0.30 rank(?width)	higher

Candidate signals:

Signal	Definition	Why test it
mom5	5-session return	shorter horizon (option 3's short mode)
mom60_skip5	return from 60 to 5 sessions ago	intermediate momentum, skipping the reversal-prone latest week
mom60_riskadj	mom60_skip5 / 60-session return ?	momentum per unit of risk
vol_5_50	5-session / 50-session average volume	multi-day volume instead of one session
squeeze_own	Bollinger width / its own 60-session median	a true squeeze: tight for that stock
low_vol	20-session return ?	lower ranks better -- tests whether squeeze is just low volatility
near_high	close / 60-session high ? 1	proximity to highs
vs_sma50, vs_sma20	distance above the 50- / 20-session SMA	trend strength

Data

- Universes: the committed universe_cache.csv snapshots from git history

(2026-04-17)

- Formation dates: every Friday session from 2026-04-17 onward that has enough forward

data for the

- Prices: yfinance daily bars, split- and dividend-adjusted, from 2025-11-03.
- Eligible at formation: price $\geq$ \$1; 20-session average dollar volume $\geq$ \$500K; snapshot
- Population ranked: eligible stocks passing the price-computable Phase 1 gates --

market ca
deal-pinne

Measurement

- Rank IC: per formation date, the Spearman correlation between a signal and the forward
- Reported per signal and horizon: mean IC, hit rate (share of dates with IC > 0), and a
- Also: the forward return of the top 50 by composite minus the population median; and the

total return
Newey-W
forward re

Decision rules -- fixed before results

Primary horizon: 10 sessions -- the default catalyst-timing directive, and it covers the

5-session

Limitations -- stated before results

- About twenty formation dates, all within five months of 2026. One market environment.
- Survivorship: stocks delisted or acquired since formation may be missing forward prices.
- The live screener computes signals on unadjusted closes; this study uses adjusted prices so
- Fundamentals cannot be tested -- no point-in-time history exists before 2026-09-14.

Coverage
splits and

Part 2 -- Results

*Run 2026-09-14. Pre-registration committed as 6c28378 (22:54 ET); the study script and its

raw output

Data as run

- 21 Friday formation dates (2026-04-17 -> 2026-09-04): 21 with 5-session forward data, 20 with
- 2,364 of 2,415 tickers priced. Forward returns exist for 99.6-100% of survivors on every
- 636-1,121 survivors per date. The universe definition changed twice during the window:
- Regime at formation: 18 dates RISK-ON, 3 RISK-OFF -- only 2 of them with 10-session data.

10, 18 with

date, so s

about ?25

Pre-registered results -- 10 sessions (primary horizon)

Signal	Mean IC	Newey-West t	IC > 0	Verdict
vol_ratio	0.052	3.22	80%	PASS
low_vol	0.113	2.88	80%	PASS
near_high	0.097	2.74	80%	PASS
vol_5_50	0.048	2.51	70%	PASS
squeeze	0.086	2.45	75%	PASS
vs_sma50	0.052	2.21	80%	PASS
composite (current)	0.072	2.15	85%	PASS
vs_sma20	0.042	1.50	65%	UNPROVEN
mom20 (40% of today's composite)	0.031	1.24	70%	UNPROVEN
mom5	0.017	0.69	55%	DROP
mom60_skip5	0.011	0.42	60%	DROP
mom60_riskadj	0.010	0.40	55%	DROP
squeeze_own	?0.007	?0.62	35%	DROP

At 5 sessions the same signals lead (low_vol t 3.42, near_high 2.96, squeeze 2.92,

vol_ratio 2

The rules applied:

- Rule 5 -- new composite: equal-weight ranks of low_vol, near_high, squeeze, vol_5_50,
- Rule 4 -- mom20: unproven, so it may keep at most its current 40%. It is removed: it adds
- Rule 7 -- timing modes: no signal passes at one horizon while ≤ 0 at the other. **One screener
- Rule 8 -- regime (descriptive): the current composite's 10-session IC was ?0.10 on the 2
- Rule 9 -- gates: every price-computable gate removed names that did worse than survivors.

vol_ratio a

nothing or

mode; opt

RISK-OFF

Median 10

A flaw in the pre-registration. Two metrics -- "top 50 minus the survivor median" and

"delivered

Post-hoc diagnostics -- not pre-registered

1. The typical stock fell; the average did not. Survivors' median 10-session return was

?0.54% (n

2. The signals mainly identify losers. The worst-ranked quintile's median 10-session return was

?3.6% for

3. With the most extreme 1% of returns clipped, the skill survives. IC against winsorized

returns at

4. Low volatility explains most of the rest. Partial IC after controlling for low_vol, at 10

sessions:

Signal	Partial IC	t
vol_5_50	0.033	2.38
vol_ratio	0.023	1.90
vs_sma50	0.033	1.72
near_high	0.039	1.61
mom20	0.016	0.80
current composite	0.011	0.61
squeeze	?0.001	?0.07
squeeze_own	?0.029	?2.47

**squeeze carries no information beyond low volatility, and neither -- nearly -- does the current

composite

5. Most of the skill appears in weeks when the median stock fell. 10-session IC when the

survivors'

6. Redundancy. Mean cross-sectional rank correlation: low_vol ~ squeeze 0.80,

near_high

7. Composites side by side (all in-sample; the rule-5 signal set was chosen on this data).

rule5_ded

10 sessions	Current	Rule 5	Rule 5 + mom20	Rule 5 dedup
Rank IC (t)	0.072 (2.15)	0.105 (2.82)	0.099 (2.70)	0.096 (3.15)
Winsorized IC (t)	0.064 (2.11)	0.085 (2.44)	0.083 (2.46)	0.083 (2.88)
Q5?Q1 median / mean	+2.5 / ?0.4pp	+3.4 / +0.3pp	+3.4 / +0.2pp	+3.3 / +0.3pp
Top 15: mean vs mean (dates +)	+0.50pp (60%)	+0.27pp (50%)	+0.34pp (65%)	+0.80pp (60%)
Top 15: median vs median (dates +)	+0.88pp (75%)	+0.91pp (70%)	+1.20pp (75%)	+0.94pp (75%)
IC, median falling / rising	0.108 / 0.005	0.169 / 0.009	0.160 / 0.009	0.137 / 0.022

At 20 sessions the top 15 beat survivors on mean by +0.38pp (current), +1.12pp (rule 5), +0.86pp

8. The watchlists the book actually researched did better than the rebuilt composite. The 18

9. Momentum. 20-day momentum -- today's largest weight -- showed no skill after gates. 60-session

Decision

Adopted -- the pre-registered outcome:

Composite: equal-weight ranks of low_vol, near_high, squeeze, vol_5_50, vol_ratio

Not adopted, tracked instead: the deduplicated composite scored best in-sample, but it was

What this means for the book. The ranking's practical edge is modest -- under 1pp per 10

Limits. 20 formation dates in one five-month period; only 2 RISK-OFF dates, while the market is

Next: Phase 3 implements the adopted composite and the shadow scores. Phase 4 re-runs this

Part 3 -- Implemented (2026-09-15)

- composite_score is now the equal-weighted percentile ranks of low_vol, near_high,

squeeze (

- New signals in screener.py: low_vol (negative 20-session return ?, so calmer ranks

higher), ne

- Shadow scores composite_legacy (the replaced 40/30/30 score) and composite_dedup (the

four-signa

- Unchanged: every gate, the sector cap, the top-50 output, the regime filter, and the rule

that scree

Open for the weekend review: portfolio_rules.md still defines momentum/technical plays as

"entry bas